

Мухаметшин А.Р. ББМО-01-25

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import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.ensemble import StackingClassifier
from sklearn.svm import SVC
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier

# -----
# Функция: визуализация данных + классификация + разделяющая поверхность
# -----
def Div_plate(clf, flip_level):
    # --- Генерация данных ---
    X, y = make_classification(
        n_samples=10000,
        n_features=2,
        n_informative=1,
        n_redundant=0,
        n_repeated=0,
        n_classes=2,
        n_clusters_per_class=1,
        class_sep=1,
        flip_y=flip_level,          # уровень шума
        weights=[0.4, 0.6],        # несбалансированные классы
        random_state=17
    )

    # --- Разделение на train/test ---
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.33, random_state=42
    )
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# --- График 1: Исходные данные ---
plt.figure(figsize=(8, 6))
ax1 = plt.gca()
sns.scatterplot(x=X_train[:, 0], y=X_train[:, 1], hue=y_train, palette='deep', ax=ax1)
sns.scatterplot(x=X_test[:, 0], y=X_test[:, 1], hue=y_test, palette='deep', ax=ax1)
ax1.set_title(f"Data With Noise (flip_y = {flip_level:.2f})")
plt.legend(title='Class', loc='upper right')
plt.show()

# --- Обучение и оценка ---
clf.fit(X_train, y_train)
y_pred = clf.predict(X_test)
score = clf.score(X_test, y_test)
print(f'Score = {score:.4f}')

# --- График 2: Правильные/неправильные предсказания ---
ind_correct = (y_test == y_pred)
ind_wrong = (y_test != y_pred)

plt.figure(figsize=(8, 6))
ax2 = plt.gca()

# Правильно классифицированные (по истинным меткам)
sns.scatterplot(x=X_test[ind_correct, 0], y=X_test[ind_correct, 1],
                hue=y_test[ind_correct], palette='deep', ax=ax2, legend=False)

# Ошибки (по предсказанным меткам)
sns.scatterplot(x=X_test[ind_wrong, 0], y=X_test[ind_wrong, 1],
                hue=y_pred[ind_wrong], palette='deep', marker='+', ax=ax2, legend=False)

ax2.set_title("Classification Results (Correct = dots, Errors = +)")
plt.legend(title='True / Predicted', loc='upper right')

# --- Разделяющая поверхность ---
plot_step = 0.02
x_min, x_max = X[:, 0].min() - 1, X[:, 0].max() + 1
y_min, y_max = X[:, 1].min() - 1, X[:, 1].max() + 1

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xx, yy = np.meshgrid(np.arange(x_min, x_max, plot_step),
                      np.arange(y_min, y_max, plot_step))

Z = clf.predict(np.c_[xx.ravel(), yy.ravel()])
Z = Z.reshape(xx.shape)

plt.contourf(xx, yy, Z, levels=1, colors=['#1f77b4', '#ff7f0e'], alpha=0.5)
plt.show()

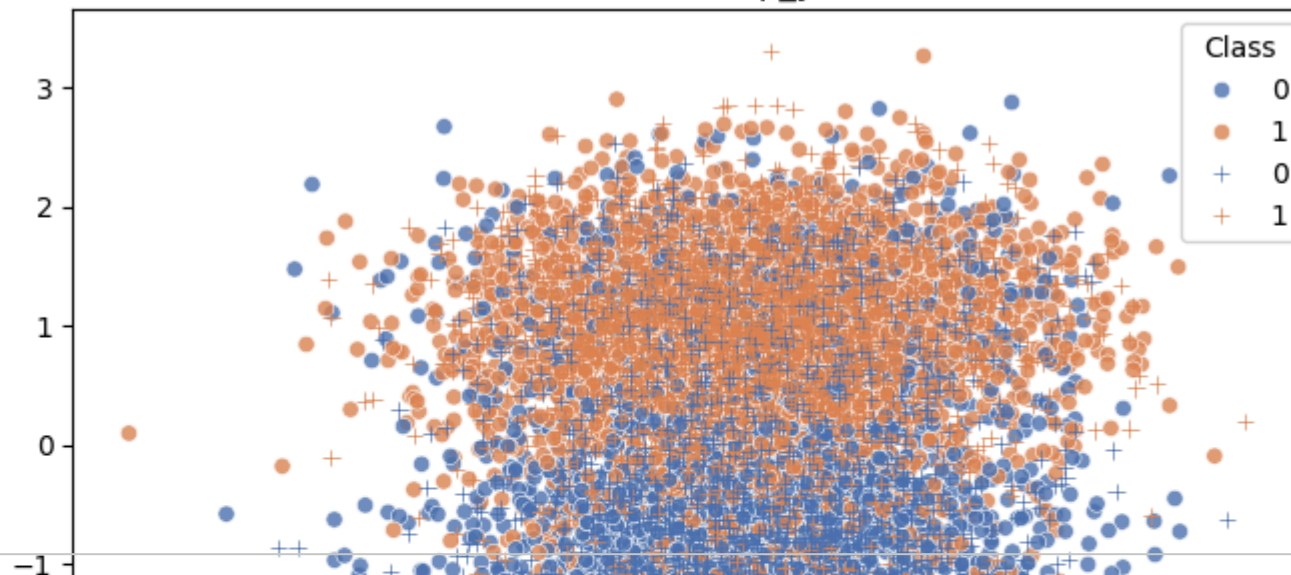
# -----
# Создание StackingClassifier
# -----
estimators = [
    ('SVC', SVC(max_iter=1000, probability=True, random_state=42)),
    ('LogReg', LogisticRegression(max_iter=1000, random_state=42)),
    ('Tree', DecisionTreeClassifier(max_depth=5, random_state=42))
]

clf = StackingClassifier(
    estimators=estimators,
    final_estimator=LogisticRegression(max_iter=1000), # по умолчанию -
    cv=5,
    passthrough=False
)

# -----
# Запуск с шумом
# -----
flip_level = 0.4
Div_plate(clf, flip_level)
```

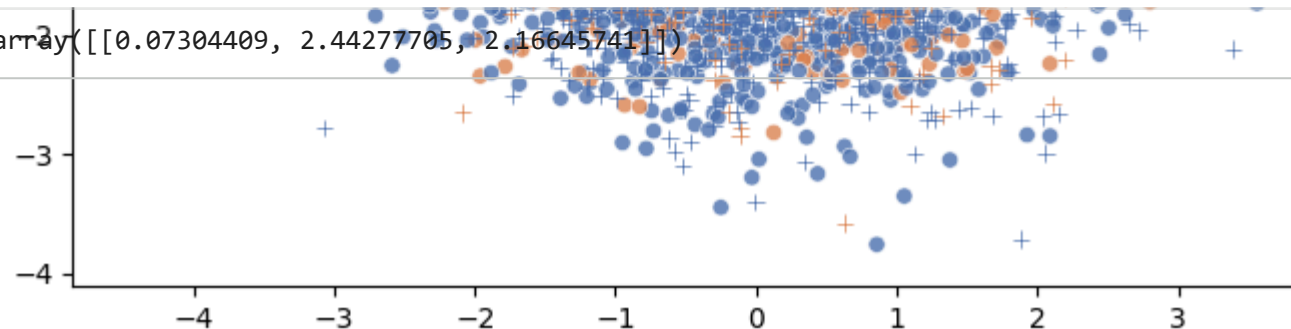


Data With Noise (flip_y = 0.40)



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clf.final_estimator_.coef_
```

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array([[0.07304409, 2.44277705, 2.16645741]])
```



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/usr/local/lib/python3.12/dist-packages/sklearn/svm/_base.py:305: ConvergenceWarning: Solver terminated early (max_iter=
warnings.warn(
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```

Score = 0.7627

Classification Results (Correct = dots, Errors = +)

4 True / Predicted